
Hidden-State Non-Recoverability and Action-Relevance Certificates for Agent Systems

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Abstract

Auditing a deployed language-model agent requires two lower certificates: one that the visible transcript cannot recover the operative hidden state, and one that the hidden state still affects the next action. We introduce a dual-certificate protocol for $\varepsilon_{\text{unrec}}^{\text{LB}}(t) \leq H(S_t \mid \tilde{T}_t)$ and $\delta_{\text{act}}^{\text{LB}}(t) \leq I(S_t; A_t \mid \tilde{T}_t)$. The first certificate is obtained from finite hidden-state witnesses or closed-form finite-dimensional witness models; the second is obtained from admissible replay, intervention, or proxy variables by conditional data processing. A structural cut-set budget remains useful, but only as an upper envelope on how much hidden state can escape the trace. The key audit failure is therefore a gap: citation-level or transcript-level audit can pass while a decision-relevant state variable remains unrecoverable. We instantiate the framework on multi-agent private-report edges, diffusion-LM denoising latents, and ReAct scratchpads. The strongest realized-action result is multi-agent: swapping a worker’s private report in a controller changes the API-action distribution by 0.901 bits (95% CI [0.873, 0.928]). LLaDA perturbations rise at the final binding step (0.110 bits, 95% CI [0.052, 0.234]). ReAct provides a conservative soft-policy diagnostic: under the same topology, controlled replay separates dormant calculator tasks from active planning tasks in tool-token probabilities (0.0163 bits, 95% CI [0.0124, 0.0208]), without changing argmax tool counts. These studies expose states that are unrecoverable from the transcript yet still action-relevant; independent state-witness non-recoverability is demonstrated in the closed-form laboratories and synthetic ground-truth calibration.

1 Introduction

Consider a controller agent that receives a worker’s private report and then chooses between a calculator and a search API. The report can pass a transcript-level audit—it need not leak the API label verbatim—while a counterfactual swap of the worker report changes the controller’s realized action distribution by 0.901 bits. This is the audit failure mode: the visible transcript can look adequate while an unobserved routing or evidence state remains both unrecoverable and decision-relevant. A weaker version appears in ReAct scratchpads: controlled replay changes the soft tool-token distribution without changing argmax tool counts. That is useful as a policy-readout diagnostic, but it is not the headline behavioral result. Identifying which hidden channels are merely present, and which ones actually drive decisions, is the audit task. How much internal state is unrecoverable from the trace, and how much of that state drives the next action, are distinct quantities. Treating them as a single “hiddenness” judgment produces something neither computable nor diagnostic: a system with unrecoverable but dormant hidden state is different from one whose unrecoverable state changes the action distribution, even when the visible transcript looks equally complete.

We propose a dual-certificate protocol that answers each question separately. The non-recoverability certificate $\varepsilon_{\text{unrec}}^{\text{LB}}(t)$ lower-bounds residual hidden-state uncertainty $H(S_t | \tilde{T}_t)$ by measuring finite witnesses of the hidden state or by evaluating closed-form finite-dimensional witness models. The action-relevance certificate $\delta_{\text{act}}^{\text{LB}}(t)$ lower-bounds $I(S_t; A_t | \tilde{T}_t)$ from admissible replay, intervention, or proxy measurements using conditional data processing. A structural cut-set budget $\varepsilon_{\text{state}}^{\text{UB}}$ is still reported when architecture access is available, but its role is an upper envelope on residual capacity, not the main NeurIPS contribution. The central audit question is whether *both*

$$H(S_t | \tilde{T}_t) \geq \varepsilon_{\text{unrec}}^{\text{LB}}(t), \quad I(S_t; A_t | \tilde{T}_t) \geq \delta_{\text{act}}^{\text{LB}}(t)$$

hold with nontrivial lower bounds. In real systems where no direct finite witness of S_t is exposed, a nonzero $\delta_{\text{act}}^{\text{LB}}$ still certifies non-recoverability through the elementary bound $I(S_t; A_t | \tilde{T}_t) \leq H(S_t | \tilde{T}_t)$. We call this an *action-mediated* non-recoverability certificate. It is weaker than an independent state-only witness, but it is exactly the gap an audit cares about: hidden state omitted from the transcript changes the action.

The dynamic axis can also be indexed. Instead of asking only whether residual state matters, an auditor can ask *where* it matters: which module, denoising step, or communication edge carries decision relevance. This turns the certificate from a scalar diagnostic into an activation profile over the hidden-state geometry of the deployment.

Conditioning on $\tilde{T}_t$ is part of the protection, not a nuisance term. If a model writes a steganographic or compressed rationale into the recorded trace, that information is in $\tilde{T}_t$ and is conditioned away by $I(S_t; A_t | \tilde{T}_t)$. The dynamic certificate measures residual decision relevance beyond the logged trace; it does not count hiddenness that has already been made visible to the audit record.

Contributions. First, we separate transcript recoverability from residual action relevance as two distinct audit questions. Second, we give dual lower certificates $(\varepsilon_{\text{unrec}}^{\text{LB}}, \delta_{\text{act}}^{\text{LB}})$ for those questions. Third, under explicit audit-abstraction assumptions, we prove certificate validity: finite hidden-state witnesses lower-bound $H(S_t | \tilde{T}_t)$, and admissible probes lower-bound $I(S_t; A_t | \tilde{T}_t)$. Fourth, we give closed-form or computable audit instances for diffusion trajectories, retrieval-style hidden source selection, and nested agent architectures. An auxiliary cut-set budget is used only for logging ablations and deployment reporting.

We instantiate the protocol on three audit geometries: a ReAct scratchpad module, a diffusion-LM denoising trajectory, and a multi-agent private-report edge. These real-system diagnostics instantiate the structural budget and action-mediated lower certificate. Appendix calibrations cover a synthetic ground-truth setting where an independent hidden-state witness makes $\varepsilon_{\text{unrec}}^{\text{LB}}$ explicit and the structural budget matches true hidden-state entropy.

2 Related Work

Probing, patching, and causal abstraction. Causal abstraction [Geiger et al., 2021], latent-knowledge elicitation [Burns et al., 2022], nullspace projection [Ravfogel et al., 2020], and amnesic probing [Elazar et al., 2021] are natural sources of finite-strength probe measurements: when they expose a probe variable $Z_t = \varphi(S_t)$ and the required conditional assumptions hold, its MI with action supplies a valid dynamic lower-bound term. Causal tracing/ROME [Meng et al., 2022], representation engineering [Zou et al., 2023], and activation addition [Turner et al., 2023] provide perturbation-based evidence when the patch affects action only through S_t .

Black-box audit and network information theory. Property testing [Goldreich et al., 1998] and black-box safety auditing [Casper et al., 2024, Su et al., 2024] study what can be inferred from outputs alone. The distinction here is that output-only access can support behavioral tests, but it does not by itself provide a lower bound on δ_{act} ; $\varepsilon_{\text{state}}^{\text{UB}}$ remains computable from topology when structural access is available. The structural budget proof applies a cut-set upper bound [El Gamal and Kim, 2011] to the time-unrolled DAG (full derivation in Appendix A).

Diffusion language-model agents. LLaDA is a large-scale masked diffusion language model with bidirectional denoising and instruction-following [Nie et al., 2025]. Recent agent work further studies diffusion LMs in multi-step decision and tool-use workflows, including matched comparisons to

autoregressive agents [Zhen et al., 2026]. Intermediate denoising latents in these systems are a natural hidden channel for the dynamic certificate.

3 Setup and Audit Regime

Standard information-theoretic notation follows Cover and Thomas [2006]. All logarithms are base 2; entropy and mutual information are reported in bits.

At step t : *full trace* T_t (all intermediate activations), *recorded trace* $\tilde{T}_t \subseteq T_t$ (auditor-visible), *operative state* S_t (internal computation), *action readout* A_t (the declared audit target: realized action, next token, or pre-sampling policy distribution), *unrecorded sources* U_t (exogenous inputs not in $\tilde{T}_t$).

3.1 Two Core Audit Quantities

Definition 1 (Core Audit Quantities). *The two quantities targeted by the dual certificate are:*

- **Hidden-state non-recoverability:** $\varepsilon_{\text{unrec}} := H(S_t \mid \tilde{T}_t)$, the residual entropy of the operative state given the visible trace.
- **Residual decision relevance:** $\delta_{\text{act}} := I(S_t; A_t \mid \tilde{T}_t)$, the mutual information between operative state and the declared action readout given the visible trace.

Neither quantity is directly computable in deployment: $\varepsilon_{\text{unrec}}$ requires recovering the full internal-state distribution, and δ_{act} requires observing S_t . The dual-certificate framework targets:

- a non-recoverability lower bound $\varepsilon_{\text{unrec}}^{\text{LB}} \leq \varepsilon_{\text{unrec}}$, computed from hidden-state witnesses or closed-form finite-dimensional models (§3.3);
- an empirical lower bound $\delta_{\text{act}}^{\text{LB}} \leq \delta_{\text{act}}$, estimated by admissible replay, intervention, or proxy probes and conditional data processing (§6).

The reported audit pair is $(\varepsilon_{\text{unrec}}^{\text{LB}}, \delta_{\text{act}}^{\text{LB}})$. When architecture access is available, we also report a structural budget $\varepsilon_{\text{state}}^{\text{UB}} \geq H(S_t \mid \tilde{T}_t)$ (§5).

Neither quantity is identifiable from behavioral observations alone: different causal graphs can produce identical joint distributions over $(\tilde{T}_t, A_t)$ pairs, so observing outputs does not fix $H(S_t \mid \tilde{T}_t)$ or $I(S_t; A_t \mid \tilde{T}_t)$. The non-recoverability certificate addresses the first by measuring or analytically computing hidden-state witnesses; the action-relevance certificate addresses the second by measuring variables whose influence on action must pass through the operative state. The structural budget bounds how large the first quantity could be under the deployment topology.

3.2 Audit Abstraction Assumptions

The certificates are claims about an audit abstraction, not about arbitrary deployment internals. The abstraction declares the variables the auditor will treat as state, transcript, action, witnesses, probes, and optional cut channels.

Assumption 1 (Deployment abstraction). *For the audited step t , the deployment induces a joint distribution over $(S_t, \tilde{T}_t, A_t)$ at a declared finite audit resolution. Any continuous internal variable used in a certificate is first mapped to a finite-noise, finite-dimensional witness or probe.*

Assumption 2 (Witness and probe admissibility). *Each non-recoverability witness $W_t^{(j)}$ is generated from S_t by a known channel with $W_t^{(j)} \perp\!\!\!\perp \tilde{T}_t \mid S_t$. Each action-relevance probe $X_t^{(j)}$ affects the next action only through the operative state after conditioning on the transcript: $X_t^{(j)} \rightarrow S_t \rightarrow A_t$ given $\tilde{T}_t$.*

Assumption 3 (Admissible cut budget). *When a structural upper budget is reported, the deployment abstraction also provides a time-unrolled graph whose unlogged cuts separate unrecorded sources from S_t , together with per-edge capacity budgets that upper-bound the conditional information crossing each unlogged channel.*

Theorem 1 (Dual certificate validity). *Under Assumptions 1–2, let $\varepsilon_{\text{unrec}}^{\text{LB}}(t)$ be any lower confidence bound on $\max_j I(W_t^{(j)}; S_t \mid \tilde{T}_t)$, and let $\delta_{\text{act}}^{\text{LB}}(t)$ be any lower confidence bound on $\max_j I(X_t^{(j)}; A_t \mid \tilde{T}_t)$. Then*

$$H(S_t \mid \tilde{T}_t) \geq \varepsilon_{\text{unrec}}^{\text{LB}}(t), \quad I(S_t; A_t \mid \tilde{T}_t) \geq \delta_{\text{act}}^{\text{LB}}(t).$$

Thus the certified state is unrecoverable from the transcript, yet still action-relevant whenever both lower bounds are nonzero.

Proof. The first inequality follows from the witness certificate (Proposition 1) applied to each $W_t^{(j)}$ and then maximized over j . The second follows from conditional data processing (Proposition 2) applied to each admissible probe and then maximized over j . Replacing population quantities by valid lower confidence bounds preserves both inequalities. $\square$

Remark 1 (Complementarity and certificate directions). *By the data processing inequality, $\delta_{\text{act}} \leq \varepsilon_{\text{unrec}}$. The reported lower certificates remain independent axes of audit evidence:*

- If $\varepsilon_{\text{state}}^{\text{UB}} = 0$ (fully logged architecture), then $\varepsilon_{\text{unrec}} = 0$ and $\delta_{\text{act}} = 0$; any valid lower bound satisfies $\delta_{\text{act}}^{\text{LB}} = 0$.
- If $\delta_{\text{act}}^{\text{LB}} = 0$ under the available admissible probes, unrecoverable state may still exist: the available witnesses may expose $\varepsilon_{\text{unrec}}^{\text{LB}} > 0$ while the current task does not activate that hidden capacity. Conversely, $\delta_{\text{act}}^{\text{LB}} > 0$ proves that the transcript omitted decision-relevant state, even if no separate non-recoverability witness has been measured.

The audit interpretation is therefore two-dimensional: deployment teams should report both lower certificates and, when possible, the structural upper budget, rather than collapse them into a single hiddenness score.

3.3 Non-Recoverability Certificate

Remark 2 (Finite-precision state entropy). *Agent states may be continuous, high-dimensional, or infinite-dimensional. Throughout the main certificate statements, $H(S_t \mid \tilde{T}_t)$ denotes audit entropy at a declared finite resolution: equivalently, the entropy of a finite partition or finite-precision witness generated by S_t . This avoids using coordinate-dependent differential entropy as an audit claim. Any continuous-state calculation below enters only after projecting to a finite-dimensional, finite-noise witness.*

Proposition 1 (Hidden-state witness certificate). *Let W_t be a finite witness generated from the operative state through a known channel $K_W(w \mid S_t)$, with $W_t \perp\!\!\!\perp \tilde{T}_t \mid S_t$. Then*

$$I(W_t; S_t \mid \tilde{T}_t) = H(W_t \mid \tilde{T}_t) - H(W_t \mid S_t) \leq H(S_t \mid \tilde{T}_t).$$

Consequently any lower confidence bound on $I(W_t; S_t \mid \tilde{T}_t)$ is a valid non-recoverability certificate $\varepsilon_{\text{unrec}}^{\text{LB}}$. If $W_t = \phi(S_t)$ is deterministic, this reduces to $H(W_t \mid \tilde{T}_t) \leq H(S_t \mid \tilde{T}_t)$.

Proof. The conditional independence assumption gives $H(W_t \mid S_t, \tilde{T}_t) = H(W_t \mid S_t)$. The first equality follows from the definition of conditional mutual information. The inequality follows because $I(W_t; S_t \mid \tilde{T}_t) \leq H(S_t \mid \tilde{T}_t)$. $\square$

Proposition 2 (Conditional DPI dynamic certificate). *Let X_t be an admissible probe variable such that, conditional on the recorded trace $\tilde{T}_t$, its influence on the next action is mediated by the operative state:*

$$X_t \longrightarrow S_t \longrightarrow A_t \quad \text{given } \tilde{T}_t.$$

Then

$$I(X_t; A_t \mid \tilde{T}_t) \leq I(S_t; A_t \mid \tilde{T}_t) = \delta_{\text{act}}.$$

Consequently any lower confidence bound on $I(X_t; A_t \mid \tilde{T}_t)$ is a valid dynamic certificate $\delta_{\text{act}}^{\text{LB}}$.

Proof. This is the conditional data processing inequality applied on each recorded trace slice and then averaged over $\tilde{T}_t$. $\square$

Corollary 1 (Action-mediated non-recoverability). *Any valid action-relevance certificate is also a valid lower bound on transcript non-recoverability:*

$$\delta_{act}^{LB} \leq I(S_t; A_t \mid \tilde{T}_t) \leq H(S_t \mid \tilde{T}_t).$$

Equivalently, $\varepsilon_{unrec,act}^{LB} := \delta_{act}^{LB}$ is a conservative non-recoverability lower bound mediated by the action channel. Thus a positive replay, intervention, or proxy certificate proves that the visible transcript omits at least that many bits of action-relevant hidden state. This certificate is action-mediated; it should not be confused with an independent state-only witness certificate from Proposition 1.

Proof. The first inequality is Proposition 2. The second is the standard bound $I(S_t; A_t \mid \tilde{T}_t) \leq H(S_t \mid \tilde{T}_t)$. $\square$

3.4 Closed-Form Witness Models

The audit interface only needs finite variables: witnesses of hidden state and probes of action relevance. Closed-form models are useful when they generate those variables with declared noise and resolution; Propositions 1 and 2 then turn the resulting information rates into audit certificates.

Let $\{P_j\}$ be finite-dimensional projections onto auditable hidden coordinates (module state, denoising step, retrieval mixture, routing layer). A Gaussian finite-precision witness has the form

$$W_t^{(j)} = P_j S_t + \eta_j Z_j, \quad Z_j \sim \mathcal{N}(0, I),$$

with known audit noise $\eta_j > 0$. The displayed Gaussian quantity is the finite-noise information in the projected hidden state beyond the trace; any subsequent finite-bin quantization can only decrease information, so lower confidence bounds remain valid finite-resolution audit certificates. In a linear Gaussian state-space model, the posterior covariance $\Sigma_{j|\tilde{T}_t} := \text{Cov}(P_j S_t \mid \tilde{T}_t)$ gives the closed-form non-recoverability certificate

$$\varepsilon_{unrec,j}^{LB}(t) = \frac{1}{2} \log_2 \det(I + \eta_j^{-2} \Sigma_{j|\tilde{T}_t}).$$

If the action readout on the same coordinate is $A_t = B_j P_j S_t + \zeta_t$ with $\zeta_t \sim \mathcal{N}(0, R_A)$, the matching action-relevance term is

$$\delta_{act,j}^{LB}(t) = \frac{1}{2} \log_2 \det(I + R_A^{-1} B_j \Sigma_{j|\tilde{T}_t} B_j^\top),$$

or any lower confidence bound on an intervention/proxy estimate of the same conditional information.

Proposition 3 (Finite-witness aggregation certificates). *For any finite set of witnesses $\{W_t^{(j)}\}_{j \in J}$ and admissible action probes $\{X_t^{(j)}\}_{j \in J}$,*

$$\max_{j \in J} I(W_t^{(j)}; S_t \mid \tilde{T}_t) \leq H(S_t \mid \tilde{T}_t), \quad \max_{j \in J} I(X_t^{(j)}; A_t \mid \tilde{T}_t) \leq I(S_t; A_t \mid \tilde{T}_t).$$

Thus any finite witness family can generate both $\varepsilon_{unrec}^{LB}(t)$ and $\delta_{act}^{LB}(t)$ as coordinate-wise lower certificates.

Proof. The first inequality is Proposition 1 applied to each witness and then maximized. The second is Proposition 2 applied to each admissible probe and then maximized. $\square$

3.5 Audit Access Modes

1. *Structural access.* The auditor has the deployment architecture, logging manifest, and interface budgets. This is sufficient to compute ε_{state}^{UB} from the unlogged subgraph (§5) but gives no lower certificate by itself.
2. *Witness access.* The auditor can read or synthesize a finite-precision witness W_t generated by S_t , or can evaluate a finite-dimensional witness model. This supports ε_{unrec}^{LB} through Proposition 1.
3. *Counterfactual replay access.* The auditor can rerun the same system under wild and replayed missing-state conditions while holding the visible trace fixed. This supports replay variables R_t for Proposition 2.

4. *Proxy or intervention access.* The auditor can read a proxy $Z_t = f(S_t)$ or perturb a hidden module with an exogenous intervention ξ_t . These support dynamic certificates when the conditional Markov condition in Proposition 2 is justified by the protocol.

These modes are instantiated in the controlled experiments below. Paraphrase- or format-based black-box replay is not used as evidence for internal decision relevance.

4 Closed-Form Audit Laboratories

The certificate interface above is intentionally small. To use it in a machine-learning paper, the mathematical backend must produce quantities that look like audit claims. This section records three analytically tractable laboratories where the same two lower bounds are computable.

Diffusion and score-flow agents. Let S_k be a denoising latent, $\tilde{T}_k = C_k S_k + \nu_k$ the logged observation, and $A_k = B_k S_k + \zeta_k$ the tool or token readout. For a linear Gaussian Markov flow

$$S_{k+1} = F_k S_k + \xi_k, \quad \xi_k \sim \mathcal{N}(0, Q_k),$$

the Kalman/Schur posterior covariance $\Sigma_{k|\tilde{T}} := \text{Cov}(S_k | \tilde{T}_k)$ gives, for any finite-rank projection P_j ,

$$\varepsilon_{\text{unrec},j}^{\text{LB}}(k) = \frac{1}{2} \log_2 \det(I + \eta_j^{-2} P_j \Sigma_{k|\tilde{T}} P_j^\top),$$

and the linear readout gives the corresponding $\delta_{\text{act},j}^{\text{LB}}(k)$ as in §3.4. This is the closed-form version of a temporal activation profile: early denoising can have nonzero unrecoverable latent volume while late denoising carries most of the action relevance.

Retrieval-augmented generation. Let C_t be a hidden source-selection state: document index, mixture component, or retrieval cluster. Let $\tilde{T}_t$ be the retrieved transcript exposed to the auditor and A_t the answer or API action. The finite laboratory has exact certificates

$$\varepsilon_{\text{unrec}}^{\text{LB}}(t) = H(C_t | \tilde{T}_t), \quad \delta_{\text{act}}^{\text{LB}}(t) = I(C_t; A_t | \tilde{T}_t),$$

with both terms computed from the table $p(c, \tilde{t}, a)$. This captures the RAG failure mode directly: the citation transcript can pass an audit while the answer distribution still carries conditional information about a source-selection state not recoverable from the transcript.

Nested agent architectures. A planner, router, memory updater, and executor can be written as a Markov kernel composition

$$S_t^{(0)} \rightarrow S_t^{(1)} \rightarrow \dots \rightarrow S_t^{(m)} \rightarrow A_t.$$

For each layer ℓ , a witness $W_t^{(\ell)}$ gives $I(W_t^{(\ell)}; S_t^{(\ell)} | \tilde{T}_t) \leq H(S_t^{(\ell)} | \tilde{T}_t)$, while a probe $X_t^{(\ell)}$ satisfying $X_t^{(\ell)} \rightarrow S_t^{(\ell)} \rightarrow A_t$ conditional on $\tilde{T}_t$ gives $I(X_t^{(\ell)}; A_t | \tilde{T}_t) \leq I(S_t^{(\ell)}; A_t | \tilde{T}_t)$. The audit output is a layerwise profile: external transcripts may look complete while internal routing or memory states retain nonzero decision relevance.

5 Structural Budget: Unlogged-Channel Cut Capacity

The structural budget upper-bounds $\varepsilon_{\text{unrec}} = H(S_t | \tilde{T}_t)$ from the deployment boundary. Information absent from the recorded trace but still capable of influencing S_t must cross an unlogged channel in the time-unrolled execution graph. Bounding the capacity of those channels gives an upper envelope on residual hidden-state entropy. This is useful for logging ablations and deployment reporting, but it is not a substitute for the lower certificates in §4–6.

Let $G_t = (V_t, E_t)$ be the time-unrolled execution graph at step t . Nodes are state updates, memory reads/writes, tool calls, and messages; edges are information channels. Let U_t denote unrecorded sources and let $\mathcal{C}_{\text{unlogged}}$ be the set of edge cuts separating U_t from S_t after logged edges are collapsed to zero residual capacity. Each unlogged edge e has a budget c_e read from the interface specification: token-window size, hidden dimension and audit quantization, or discrete scheduler cardinality.

Proposition 4 (Cut-set structural budget). *For any cut Ω separating the unrecorded sources from S_t , let $C_{\text{cut}}(\Omega)$ be the induced conditional capacity across the cut:*

$$C_{\text{cut}}(\Omega) = \sup_{p(X_\Omega | \tilde{T}_t)} I(X_\Omega \rightarrow Y_{\Omega^c} | \tilde{T}_t, X_{\Omega^c}).$$

Then

$$H(S_t | \tilde{T}_t) \leq H(S_t | T_t) + \min_{\Omega \in \text{Cuts}(U_t \rightarrow S_t)} C_{\text{cut}}(\Omega).$$

Setting the right-hand side to $\varepsilon_{\text{state}}^{UB}$ gives $\varepsilon_{\text{unrec}} \leq \varepsilon_{\text{state}}^{UB}$.

Proof. Let $M_t = T_t \setminus \tilde{T}_t$ be the unrecorded trace fragment. The chain rule gives

$$H(S_t | \tilde{T}_t) = H(S_t | T_t) + I(S_t; M_t | \tilde{T}_t).$$

For any cut Ω , every directed path from U_t or M_t to S_t crosses the cut, so conditional data processing bounds $I(S_t; M_t | \tilde{T}_t)$ by the induced information flow across that cut. Taking the supremum over cut inputs gives $C_{\text{cut}}(\Omega)$, and minimizing over cuts gives the claim. Appendix A spells out the d -separation step. $\square$

Lemma 1 (Additive decomposition under software orthogonality). *If the channel laws across a cut factor into independent single-edge channels conditional on $\tilde{T}_t$, and if c_e upper-bounds the conditional information that can cross edge e , then*

$$C_{\text{cut}}(\Omega) \leq \sum_{e \in E(\Omega, \Omega^c)} c_e.$$

Proof. Under conditional factorization, the directed conditional information across the cut is subadditive over its edge streams; substituting each edge budget c_e gives the bound. $\square$

Corollary 2 (Additive min-cut structural budget). *Under software orthogonality,*

$$\varepsilon_{\text{state}}^{UB} = H(S_t | T_t) + \min_{C \in \mathcal{C}_{\text{unlogged}}} \sum_{e \in C} c_e.$$

Corollary 3 (Autoregressive zero cut). *If the deployment is a strict autoregressive core and $\tilde{T}_t$ contains the full context window relevant to S_t , then the unlogged cut has zero capacity. If additionally $H(S_t | T_t) = 0$, then $\varepsilon_{\text{state}}^{UB} = 0$ and $\delta_{\text{act}} = 0$.*

6 Dynamic Certificate: Conditional DPI Probes

The dynamic certificate targets $\delta_{\text{act}} := I(S_t; A_t | \tilde{T}_t)$. Because S_t is unobservable, the auditor cannot measure $I(S_t; A_t | \tilde{T}_t)$ directly. Proposition 2 gives the certificate rule: measure an admissible variable whose path to the action is mediated by S_t , then use conditional data processing.

Replay certificate. Let $R_t \in \{\text{wild}, \text{replay}\}$ indicate whether a missing state fragment is present or removed under the same recorded trace. If the replay protocol affects the action only through the missing operative state, then $R_t \rightarrow S_t \rightarrow A_t$ conditional on $\tilde{T}_t$. For a balanced binary replay, $I(R_t; A_t | \tilde{T}_t)$ is the conditional JS divergence between the wild and replay action distributions.

Intervention certificate. Let ξ_t be an exogenous perturbation applied to a hidden module while holding $\tilde{T}_t$ fixed. When the protocol ensures $\xi_t \rightarrow S_t \rightarrow A_t$ conditional on $\tilde{T}_t$, the measured conditional action-distribution shift is a lower bound on δ_{act} .

Proxy certificate. Let $Z_t = f(S_t)$ be a readable coarse statistic of the operative state. Since Z_t is a function or coarsening of S_t , $I(Z_t; A_t | \tilde{T}_t) \leq \delta_{\text{act}}$. In experiments we estimate this term by cross-fitted cross-entropy differences between predictors with and without Z_t .

Logging level	$\varepsilon_{\text{state}}^{\text{UB}}$ (bits)	Bottleneck
ReAct, output only	16,464	scratchpad_read (16384) + router_logits (80)
ReAct, +log router	16,384	scratchpad_read (16384)
ReAct, +log scratchpad	0	none
ReAct, full instrumentation	0	none
LLaDA-8B-Instruct ($K=128$)	32,768	denoising interface
Multi-agent, all logged	0	none
Multi-agent, peer report unlogged	32,768	peer_report $\rightarrow S_t$

Table 1: Structural budget geometry. Adding ReAct logs monotonically removes residual cut capacity; the same min-cut analysis localizes capacity in a scratchpad module, denoising interface, or peer-report edge. The LLaDA bound is independent of K because repeated reuse of the same bounded latent interface does not increase the one-step information bottleneck.

Aggregation. For a set $\mathcal{X}$ of admissible probe variables, define

$$\delta_{\text{act}}^{\text{LB}} = \max_{X_t \in \mathcal{X}} \hat{I}_{\text{LCB}}(X_t; A_t \mid \tilde{T}_t),$$

where $\hat{I}_{\text{LCB}}$ is the reported lower confidence bound. The maximum remains a lower bound because every admissible term is bounded above by δ_{act} .

Activation profiles. Indexing admissible probes over hidden-channel coordinates (module, denoising step, communication edge) gives an *activation profile*

$$\Delta(j) := \hat{I}_{\text{LCB}}(X_t^{(j)}; A_t \mid \tilde{T}_t),$$

where j denotes the coordinate being probed. The structural budget identifies where residual capacity can reside; the activation profile identifies where that capacity is behaviorally used.

7 Empirical Diagnostics

The real-system experiments do not expose an independent state-only witness W_t for S_t . They therefore report two quantities: a structural budget $\varepsilon_{\text{state}}^{\text{UB}}$ from topology, and an action-relevance lower certificate $\delta_{\text{act}}^{\text{LB}}$. By Corollary 1, every positive $\delta_{\text{act}}^{\text{LB}}$ is also an action-mediated lower bound on $H(S_t \mid \tilde{T}_t)$. ReAct intervention and controlled replay (§7.2) show the same unlogged scratchpad dormant on calculator tasks and active on planning tasks. The LLaDA and multi-agent probes index $\Delta(j)$ by denoising step and peer-report edge, respectively (§7.3–7.4). Independent state-witness calibration is provided by the synthetic ground-truth experiment in Appendix D. All pipelines run on a single M4 Max workstation.

7.1 Structural Budget: Min-Cut and Logging Ablation

We extract the time-unrolled DAG from Qwen2.5-7B-Instruct ReAct execution traces on tool-selection tasks (Fig. 1, left). Per-edge budgets c_e are assigned from interface specifications (§5); the min-cut on unlogged edges gives $\varepsilon_{\text{state}}^{\text{UB}}$. Varying logging levels recomputes the min-cut; Table 1 reports the stepwise reduction from 16,464 to 0 bits. The dominant bottleneck is the scratchpad-read path (16,384 bits). The same structural computation also supplies the coordinate system for the diffusion-LM and multi-agent dynamic checks that follow.

These cuts put the later dynamic probes on a common map: a scratchpad module in ReAct, a denoising interface in LLaDA, and a peer-state edge in the multi-agent system.

7.2 Dynamic Causal Certificates: Intervention and Controlled Replay

We separate the two axes with a dormant/active task split under one topology. The unlogged scratchpad is present in both cases: irrelevant for calculator tasks, necessary for planning tasks. Intervention perturbs the scratchpad (masking or replacement); controlled replay removes it entirely

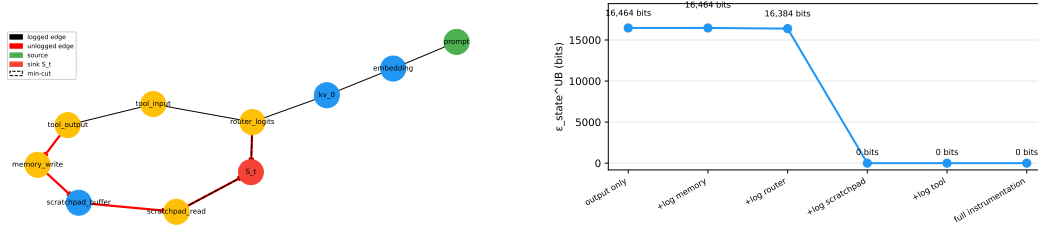


Figure 1: Left: Time-unrolled DAG extracted from Qwen2.5-7B-Instruct ReAct execution (logical edges omitted for clarity). Red edges are unlogged. Dashed edges form the min-cut. Right: Logging ablation curve. Adding instrumentation at the bottleneck drops $\epsilon_{\text{state}}^{\text{UB}}$ stepwise to zero.

Task	Perturbation	JS (bits; nats)	95% CI (bits)
Calculator (dormant)	mask 0.7	0.0000; 0.0000	[0, 0]
Planning (active)	mask 0.7	0.0188; 0.0131	[0.0070, 0.0381]
Calculator (dormant)	replace 1.0	0.0000; 0.0000	[0, 0]
Planning (active)	replace 1.0	0.0080; 0.0056	[0.0027, 0.0167]

Table 2: Dynamic intervention certificate under the same unlogged-scratchpad topology. Dormant calculator tasks show no detectable scratchpad influence; active planning tasks show positive action-distribution shift.

while holding $\tilde{T}_t$ fixed. Table 2 reports JS divergence over realized tool actions. Table 3 reports JS divergence over the model’s soft tool-token probability distribution; for the policy-readout replay contrast, the argmax tool counts are unchanged, so the evidence is a policy-probability shift rather than a realized-tool flip. Both action-relevance certificates are zero on dormant tasks and positive on active tasks. The topology, and therefore the structural budget $\epsilon_{\text{state}}^{\text{UB}}$, is unchanged.

The pair of tables isolates the transition from high structural budget with zero action-mediated lower certificate to high structural budget with positive action-mediated lower certificate: the hidden channel is structurally present in both task splits, but it becomes action-relevant only on planning tasks.

Under this fixed topology, the unlogged scratchpad contributes 16,384 bits to $\epsilon_{\text{state}}^{\text{UB}}$ in both task splits, matching the min-cut from the extracted ReAct DAG. Intervention and replay are zero on calculator tasks and positive on planning tasks. The structural budget is set by the deployment boundary and logging policy; residual decision relevance depends on which hidden channels the task uses. The positive planning rows are also action-mediated non-recoverability certificates by Corollary 1; they are not independent state-only witness certificates. The controlled replay row should be read as a pre-sampling policy-readout certificate, not as a realized-action flip.

7.3 Temporal Certificate Analysis on a Diffusion LM

ReAct gives a module-level contrast: the scratchpad is either used or not used. A diffusion LM gives a temporal contrast. The hidden channel exists throughout the K -step denoising trajectory, but its decision relevance need not be constant across steps.

We use LLaDA-8B-Instruct with $K=10$ denoising steps on tool-selection prompts. A Gaussian perturbation ($\sigma=5.0$) is applied to layer-1 activations at steps $\{2, 4, 6, 8, 10\}$; a late-layer perturbation is tracked as a broad sensitivity control. Each point uses $n=20$ trajectories and a trajectory-block bootstrap with 1,000 resamples.

The pattern is late-binding. Steps 2–8 remain near the floor for the target layer (0.001–0.006 bits), while step 10 rises to 0.110 bits with 95% CI [0.052, 0.234]. The control layer also rises at step 10 (0.092 bits, CI [0.043, 0.186]), so the sensitivity is global rather than confined to the chosen layer. LLaDA carries hidden capacity throughout denoising; it becomes action-relevant when the denoising state binds to a discrete tool choice. The step-10 value is therefore an action-mediated non-recoverability lower certificate, not a direct entropy estimate of the full denoising latent.

Task	Replay	JS (bits; nats)	95% CI	Argmax
Calc. (dormant)	removed	0.0000; 0.0000	[0, 0]	same
Planning (active)	removed	0.0163; 0.0113	[0.0124, 0.0208]	same

Table 3: Controlled replay certificate over soft tool-token probability distributions. Wild episodes use the full hidden scratchpad; replay episodes use the same visible trace with the scratchpad removed. The dormant-to-active contrast mirrors the intervention result at the soft-policy level. In both rows, aggregate argmax tool counts are unchanged.

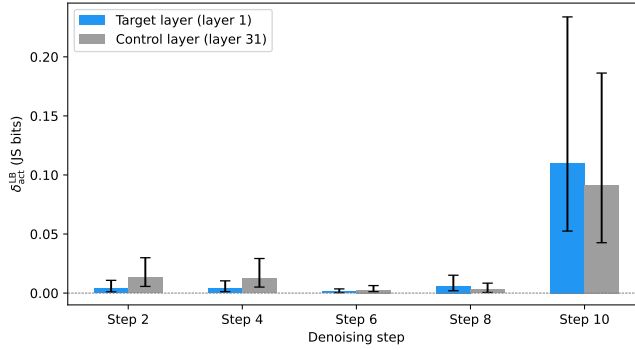


Figure 2: Temporal activation profile for LLaDA-8B-Instruct. Early denoising steps have low dynamic certificates; the final action-forming step has a much larger certificate for both target and control layers, indicating late-binding decision relevance rather than a module-local scratchpad effect.

7.4 Multi-Agent Private-Report Intervention

The multi-agent setting changes the index again. Here the controller delegates evidence gathering to a worker, so the worker’s private report is not an auxiliary scratchpad but an epistemic edge in the decision graph. The worker sees a benchmark item and writes a short report; the controller sees only that report and chooses one API from {calculator, search}. The task pool combines 200 GSM8K arithmetic items [Cobbe et al., 2021] and 200 HotpotQA multi-hop fact questions [Yang et al., 2018] with seed 42.

The main intervention replaces the worker report with a report from the opposite task class while holding the controller protocol fixed. A neutral replay removes the report content, and a same-class shuffle replaces it with another report from the same task class. An oracle-tag condition directly names and flips the API label to measure the binary ceiling. For each task and condition, the controller is sampled $k=8$ times; labels are parsed as {calculator, search, invalid}; and $JS(P_{base}, P_{condition})$ is averaged with a task-block bootstrap.

The contrast is topological. In ReAct, the scratchpad matters only when the task needs it. In the multi-agent deployment, the controller’s decision is routed through the private worker report, so the active site is the communication edge itself. Same-class variation is local, neutral removal is substantial, and counterfactual evidence reaches 0.901 bits, close to the oracle ceiling. This is again an action-mediated lower certificate: it shows that transcript-omitted private report state remains decision-relevant, while the structural budget reports how much private edge capacity could be hidden.

8 Discussion

8.1 One Plot: Capacity, Then Activation

The three real-system experiments follow a single logic. Structural budgets map where hidden capacity can reside—scratchpad modules, denoising interfaces, peer-state edges. Dynamic certificates identify which of those sites affect actions. By Corollary 1, every positive dynamic certificate is also a lower bound on transcript non-recoverability, but only through the action channel. The same

Condition	$\delta_{\text{act}}^{\text{LB}}$ (JS bits)	95% CI (bits)
Neutral replay	0.358	[0.324, 0.392]
Counterfactual evidence	0.901	[0.873, 0.928]
Same-class shuffle	0.084	[0.060, 0.109]
Oracle tag upper-bound	1.000	[1.000, 1.000]

Table 4: Topological activation profile for the multi-agent deployment. Counterfactual evidence nearly saturates the binary action ceiling, while same-class shuffling remains much smaller. Hidden communication is therefore behaviorally active when the controller has delegated evidence gathering to the worker.

scratchpad is dormant on calculator tasks and active on planning tasks in ReAct; in LLaDA, action relevance concentrates at the final binding step; in the multi-agent deployment, delegation makes the private worker report the active edge.

Effect sizes index deployment topology. Under the same ReAct topology, calculator tasks leave the scratchpad dormant while planning tasks produce nonzero realized-action shifts under intervention (0.0188 bits, CI [0.0070, 0.0381]) and a softer policy-readout shift under controlled replay (0.0163 bits, CI [0.0124, 0.0208]). The LLaDA final binding step rises to 0.110 bits; the multi-agent counterfactual report edge rises to 0.901 bits. A hidden channel that supplements an already-determined visible trace has low $\delta_{\text{act}}^{\text{LB}}$; one that carries the decision has high $\delta_{\text{act}}^{\text{LB}}$. Under this protocol, a claim that an unobserved variable causally affects behavior must identify a structural channel in $\varepsilon_{\text{state}}^{\text{UB}}$ and a lower-bound probe in $\delta_{\text{act}}^{\text{LB}}$. A stronger claim of state-only non-recoverability requires an independent witness $\varepsilon_{\text{unrec}}^{\text{LB}}$, which we demonstrate in the closed-form laboratories and synthetic ground-truth calibration rather than in the black-box real-system diagnostics.

Acknowledgments

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A Network Information Theory Background for Proposition 4

Conditional channel capacity. For each unlogged edge e in the time-unrolled DAG G_t , the conditional capacity c_e is the maximum information that can flow through e under the worst-case distribution consistent with the system protocol. It is not an empirical average mutual information. It is a structural upper bound read from interface specifications: token window length, hidden-dimension size, quantization level, and discrete-state cardinality. Concretely, a text channel with budget K tokens over vocabulary $\mathcal{V}$ has $c_e \leq K \log |\mathcal{V}|$; a vector state of dimension d at quantization Q has $c_e \leq d \log Q$; a discrete scheduler with $|\Omega|$ states has $c_e \leq \log |\Omega|$. The $K \log |\mathcal{V}|$ form is the maximum-entropy bound (uniform over K -token sequences); any other distribution gives strictly lower entropy. Auditors who can measure the empirical input distribution to a particular channel may substitute a tighter bound, which only improves the structural budget.

What tightening means. The structural budget remains valid under any replacement $c_e^* \geq I(X_e; Y_e \mid \tilde{T}_t)$, so tightening is a matter of choosing smaller but still defensible per-edge budgets. In practice, this means replacing the worst-case $K \log |\mathcal{V}|$ or $d \log Q$ budget with an empirical or conditional estimate at the same audit discretization, rather than changing the theorem itself.

Cut-set upper bound and additive decomposition. Let $G_t = (V_t, E_t)$ be the time-unrolled DAG. For a cut $\Omega \subseteq V_t$ separating source nodes from a target node, the *induced cut capacity* $C_{\text{cut}}(\Omega)$ is the supremum, over joint input distributions $p(X_\Omega \mid \tilde{T}_t)$, of the directed conditional mutual information $I(X_\Omega \rightarrow Y_{\Omega^c} \mid \tilde{T}_t, X_{\Omega^c})$. The network-information-theoretic cut-set upper bound [El Gamal and Kim, 2011] states that any information flowing from sources in Ω to a target in Ω^c is bounded above by $C_{\text{cut}}(\Omega)$, and minimizing over cuts gives the tightest such bound. Proposition 4 applies this bound to the unrecorded trace fragment M_t . We use the cut-set *upper bound* direction only; the network-coding achievability results [Ahlsweide et al., 2000], which concern lower bounds on capacity under cooperative coding across the cut, are not invoked. The induced capacity $C_{\text{cut}}(\Omega)$ is in general a supremum over joint input distributions and cannot be read directly from interface specifications. Lemma 1 shows that under software orthogonality, when the channel laws across the cut factor into independent single-edge channels conditional on $\tilde{T}_t$, the induced capacity is bounded above by the sum of per-edge capacities:

$$C_{\text{cut}}(\Omega) \leq \sum_{e \in E(\Omega, \Omega^c)} c_e.$$

Corollary 2 combines this with Proposition 4 to reduce the structural-budget computation to a discrete minimum-cut problem on G_t with edge capacities c_e . This is the form used in the structural-budget experiment and in the protocol in §B.

Formal derivation of Proposition 4. The proof in §5 has two steps: a trace-gap identity and a cut-set bound via d -separation. We restate the argument here with the d -separation step explicit.

Step 1: Trace-gap identity. Let $M_t := T_t \setminus \tilde{T}_t$. The chain rule of conditional entropy gives

$$H(S_t \mid \tilde{T}_t) = H(S_t \mid T_t) + I(S_t; M_t \mid \tilde{T}_t).$$

The first term is $\varepsilon_{\text{nominal}} = H(S_t \mid T_t)$.

Step 2: Cut-set bound via d -separation. Fix a cut $\Omega \in \text{Cuts}(U_t \rightarrow S_t)$. By the definition of a cut in the time-unrolled DAG G_t , every directed path from a node in Ω to $S_t \in \Omega^c$ must cross at least one edge in $E(\Omega, \Omega^c)$. The causal influence of M_t on S_t must be mediated by the cross-cut message stream $(X_\Omega \rightarrow Y_{\Omega^c})$ together with the boundary state X_{Ω^c} . Formally, M_t is d -separated from S_t by $(X_\Omega \rightarrow Y_{\Omega^c}, X_{\Omega^c})$ conditional on $\tilde{T}_t$, which yields

$$I(S_t; M_t \mid \tilde{T}_t) \leq I(X_\Omega \rightarrow Y_{\Omega^c} \mid \tilde{T}_t, X_{\Omega^c}).$$

By the definition of $C_{\text{cut}}(\Omega)$ as the supremum of the right-hand side over admissible joint input distributions on X_Ω ,

$$I(S_t; M_t \mid \tilde{T}_t) \leq C_{\text{cut}}(\Omega).$$

Since this holds for every cut, it holds for the minimum,

$$I(S_t; M_t \mid \tilde{T}_t) \leq \min_{\Omega \in \text{Cuts}(U_t \rightarrow S_t)} C_{\text{cut}}(\Omega).$$

Combining with Step 1 completes the proof of Proposition 4. Corollary 2 follows immediately by substituting Lemma 1 into each software-orthogonal cut: under orthogonality, $C_{\text{cut}}(\Omega) \leq \sum_{e \in E(\Omega, \Omega^c)} c_e$, and the minimum over cuts becomes a discrete min-cut problem on the unlogged subgraph with edge capacities c_e .

B Experimental Protocols

B.1 Structural Budget Computation Protocol

Inputs: deployment architecture specification, logging manifest, and protocol budgets (token window sizes, hidden dimensions, quantization levels). **Procedure:** (1) construct the time-unrolled DAG G_t ; (2) partition edges into logged and unlogged; (3) verify software orthogonality (Lemma 1) or add shared-state edges explicitly to G_t until orthogonality holds; (4) assign capacity c_e per unlogged edge using the forms in §5; (5) enumerate reachable cut sets $\mathcal{C}_{\text{unlogged}}$ with a min-cut algorithm after logged edges are collapsed to zero residual capacity; (6) compute $\varepsilon_{\text{state}}^{\text{UB}} = \varepsilon_{\text{nominal}} + \min_C \sum_{e \in C} c_e$ via Corollary 2.

For the logging ablation, repeat steps (2)–(6) for the ordered manifests used in §7.1: output-only trace, tool-call trace, router-logit trace, summary-buffer trace, and full scratchpad trace. Report the upper-bound value and active minimum-cut edges at each level. See `experiments/7.1_static_certificate/README.md` for implementation.

B.2 Dynamic Certificate Estimation Protocol

Proxy certificate. **Inputs:** model, task environment, and probe family $Z_t^{(k)} = f_k(S_t)$. **Steps:** (1) run inference on N trajectories and record $(\tilde{T}_t, Z_t^{(k)}, A_t)$; (2) train cross-fitted predictors $p_\theta(A_t \mid \tilde{T}_t)$ and $p_\phi(A_t \mid \tilde{T}_t, Z_t^{(k)})$; (3) compute the cross-entropy difference $\hat{I}_k = L_{\text{trace}} - L_{\text{trace}+Z^{(k)}}$; (4) estimate trajectory-block bootstrap intervals; (5) repeat for random, label-permuted, 1-dimensional, 3-dimensional, 5-dimensional, and full-slice proxies.

Optional controlled replay certificate. **Inputs:** model and two controlled execution regimes (wild and replay). **Steps:** (1) for each $\tilde{T}_t$, run the same system under wild and replayed missing-state conditions; (2) estimate $\hat{\Delta} = \hat{\text{JS}}(P_{\text{wild}}, P_{\text{replay}} \mid \tilde{T}_t)$; (3) aggregate over trace slices. This is not output-only paraphrase replay; it is valid only under the controlled replay condition in §6.

Intervention certificate. **Inputs:** model, targeted hidden module, perturbation budget, and task split. **Steps:** (1) inject ξ_{hidden} at levels $\{0, \sigma_1, \sigma_2, \dots\}$ while holding $\tilde{T}_t$ fixed; (2) record $(\xi_{\text{hidden}}, A_t \mid \tilde{T}_t)$ pairs; (3) estimate $\hat{I}(\xi_{\text{hidden}}; A_t \mid \tilde{T}_t)$ via conditional action-distribution shift for discrete action spaces, or via InfoNCE/MINE for continuous action spaces; (4) repeat on dormant tasks and active tasks under the same topology to test whether $\delta_{\text{act}}^{\text{LB}}$ changes while $\varepsilon_{\text{state}}^{\text{UB}}$ remains fixed. The implementation persists per-sample argmax actions, tool-token log probabilities, and normalized tool-choice probabilities so the aggregate JS rows can be recomputed from raw intervention traces.

Multi-agent private-report certificate. **Inputs:** task pool, worker model, controller model, and a two-class API space. **Steps:** (1) sample $N/2$ arithmetic tasks and $N/2$ factual-retrieval tasks; (2) have the worker generate a private evidence report under a prompt that forbids direct API names and audit each report for leakage; (3) sample the controller action distribution k times under wild, neutral replay, opposite-class report, same-class shuffled report, and oracle-tag conditions; (4) parse actions into `{calculator, search, invalid}`; (5) compute per-task JS divergence in bits and task-block bootstrap intervals. The oracle-tag contrast is reported only as an upper bound; the main contrast is the opposite-class worker-report intervention.

See the README files for experiments 7.2, 7.3, and 7.6 for implementation.

C Proxy Estimator: Bias and Variance

The read-only proxy experiment is a calibration of observational access, not a main activation result. On 3,000 tool-selection trajectories (mixed calculator/search/email/calendar/weather), we record $\tilde{T}_t$ (embedding-layer representation), Z_t (layer-24 tool-logit projection), and A_t . The CE-diff $\hat{I} = \mathcal{L}(A_t | \tilde{T}_t) - \mathcal{L}(A_t | \tilde{T}_t, Z_t)$ is estimated via 5-fold cross-fitted logistic regression with trajectory-block bootstrap. Table 5 reports the resolution ablation: random, 1/3/5-dim PCA, and full 10-dim proxy.

The $d=5$ PCA proxy recovers 0.0317 bits (1.4% of the $\log_2 5 = 2.32$ bit ceiling) with CI excluding zero on the mixed-task distribution. The magnitude reflects an *observational bottleneck*: when the baseline predictor achieves near-perfect accuracy, the ceiling $H(A_t | \tilde{T}_t) \approx 0$ caps any read-only proxy. Intervention and replay avoid this bottleneck by changing the missing-state condition while holding the visible transcript fixed.

The dormant/active proxy split should be read as an estimator-noise diagnostic. On calculator tasks (dormant scratchpad), $H(A_t | \tilde{T}_t) = 0$ empirically, yielding $\delta_{\text{act}}^{\text{LB}} = 0$ trivially. On planning tasks (active scratchpad), the proxy row is positive in point estimate but has a wide CI that straddles zero. We therefore do not use the read-only proxy as the main body evidence for active scratchpad use; the causal intervention and controlled replay in §7.2 supply that evidence.

Proxy	CE-diff (bits; nats)	95% CI (bits)
<i>Resolution ablation (mixed task)</i>		
random	−0.0007; −0.0005	[−0.0012, −0.0003]
$d = 1$ PCA	+0.0065; +0.0045	[+0.0053, +0.0078]
$d = 3$ PCA	+0.0250; +0.0173	[+0.0225, +0.0273]
$d = 5$ PCA	+0.0317; +0.0220	[+0.0273, +0.0358]
full (10-dim)	+0.0183; +0.0127	[+0.0134, +0.0229]
<i>Dormant/active split ($d=64$ PCA, L2-regularized)</i>		
calculator (dormant)	0.0000; 0.0000	[0, 0]
planning (active)	+0.0412; +0.0286	[−0.0725, +0.1386]

Table 5: Read-only proxy calibration. Top: resolution ablation on the mixed tool-selection dataset. Bottom: dormant/active split. The active-task row is reported honestly as an estimator-noise diagnostic because its confidence interval straddles zero; causal replay/intervention supplies the body evidence for active scratchpad use. Negative random-control CE-diff is a finite-sample artifact near the noise floor and is clipped to zero when interpreted as a lower-bound certificate.

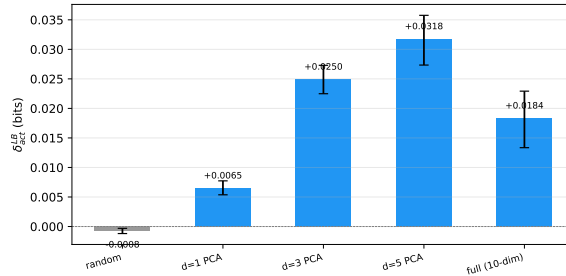


Figure 3: Proxy resolution ablation. $\delta_{\text{act}}^{\text{LB}}$ increases monotonically through $d=5$ PCA. The random proxy sits near zero (estimator leakage check). Error bars are 95% trajectory-block bootstrap CIs.

The CE-diff estimator $\hat{I} = L(\tilde{T}_t) - L(\tilde{T}_t, Z_t)$ lower-bounds $I(Z_t; A_t | \tilde{T}_t)$, but three bias sources and two variance sources matter in practice.

Bias sources. (1) *Proxy coarsening*: $Z_t = f(S_t)$ discards information in S_t not captured by f , so $I(Z_t; A_t | \tilde{T}_t) \leq I(S_t; A_t | \tilde{T}_t)$. The gap is the coarsening loss. (2) *Restricted predictor class*: Logistic regression or shallow MLPs may not reach the Bayes-optimal predictor, so the CE-diff

underestimates the true conditional MI. (3) *Finite-sample bias*: With finite N , the cross-entropy on held-out folds is an upward-biased estimate of the population CE, and the difference of two such estimates carries residual bias. Cross-fitting mitigates but does not eliminate this.

Variance sources. (1) *Trajectory sampling*: The N observed trajectories are a finite sample from the deployment distribution. Block bootstrap over trajectories estimates the resulting CI. (2) *Proxy dimension*: Higher-dimensional proxies carry more information but increase predictor variance. The PCA ablation in Table 5 shows this trade-off: $d=5$ outperforms both $d=3$ and the full 10-dimensional proxy.

Mean squared error. Decomposing $\text{MSE} = (\text{bias})^2 + \text{variance}$, the proxy certificate is a conservative lower bound: positive bias (underestimation) is safe for the audit (it never overstates δ_{act}), while variance widens the CI but does not shift the point estimate.

Sampling noise and temperature. Increasing decoding temperature can raise $H(A_t | \tilde{T}_t)$, but independent sampler noise is not a certificate amplifier. It changes the audited policy and can reduce $I(Z_t; A_t | \tilde{T}_t)$ when the added action variation is unrelated to the hidden state proxy. A temperature sweep is therefore a calibration of a randomized deployment policy, not a substitute for proxy quality or causal access. We do not use temperature injection for the main proxy certificate; the synthetic experiment instead injects hidden-state stochasticity with known ground truth.

D Synthetic Ground-Truth Validation

To test whether the certificates recover known hidden influence, we construct a synthetic agent whose hidden-state entropy and decision relevance are controlled by design. A binary hidden variable $H \sim \text{Bern}(0.5)$ injects influence of strength β_h into the action logits; the unlogged-channel capacity $c_H = H(H) = 1$ bit is known. Since H is also a deterministic hidden-state witness, the non-recoverability lower certificate satisfies $\varepsilon_{\text{unrec}}^{\text{LB}} = H(H | \tilde{T}_t) = 1$ bit. We run the structural min-cut budget ($\varepsilon_{\text{state}}^{\text{UB}} = 1$ bit), the witness certificate, and the proxy CE-diff estimator on 1,000 trajectories at five influence levels.

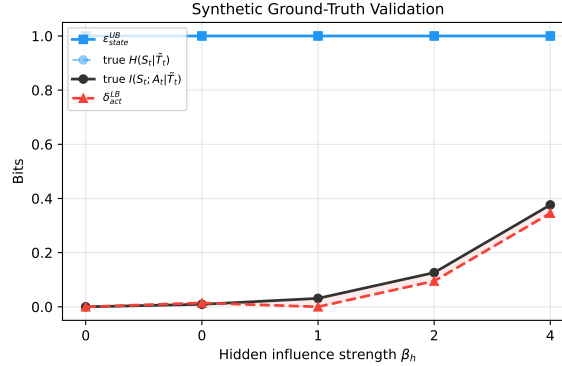


Figure 4: Synthetic ground-truth. $\varepsilon_{\text{state}}^{\text{UB}}$ (blue) and the witness lower certificate $\varepsilon_{\text{unrec}}^{\text{LB}}$ both equal the true $H(S_t | \tilde{T}_t) = 1$ bit. $\delta_{\text{act}}^{\text{LB}}$ (red) tracks the true $I(S_t; A_t | \tilde{T}_t)$ from below, with negative values at low β_h reflecting estimator variance near the noise floor.

Fig. 4 shows $\varepsilon_{\text{unrec}}^{\text{LB}} = \varepsilon_{\text{state}}^{\text{UB}} = 1$ bit, matching the true hidden entropy because the hidden variable is both the only unlogged channel and a measured witness. The dynamic lower bound $\delta_{\text{act}}^{\text{LB}}$ tracks the true conditional MI from below at all tested influence levels ($\beta_h \geq 1$); negative CE-diff values at $\beta_h = 0$ reflect finite-sample variance around a near-zero true MI.

The Qwen deployment has a near-deterministic baseline that suppresses the proxy signal ($H(A_t | \tilde{T}_t) \approx 0$). The synthetic environment instead injects controlled stochasticity ($H = 1$ bit of hidden-state entropy). Without that observational bottleneck, the proxy certificate $\delta_{\text{act}}^{\text{LB}}$ tracks the true conditional-MI curve from below in the tested range. The validation shows that the proxy certificate

can become tight when structural capacity is behaviorally active *and* the observational ceiling permits detection.

E Tightening the Structural Budget

The reported structural budget is intentionally conservative. Its role is to upper-bound the residual hidden-state entropy from deployment topology alone, so the default budgets in §5 are worst-case interface capacities rather than empirical information estimates.

Empirical and conditional budgets. A tighter deployment-specific budget can replace the worst-case budget c_e on each unlogged edge by an empirically justified budget c_e^* . For a text channel, one option is a high-confidence upper bound on $H(X_e \mid \tilde{T}_t)$; when the visible history strongly constrains the missing input, a conditional form such as $H(X_e \mid \tilde{T}_t, \mathcal{H}_t)$ can be smaller still. For quantized vector states, the analogous replacement is an empirical entropy or rate-distortion budget at the same audit discretization. These substitutions do not alter the min-cut proof: they only change the edge weights used by Corollary 2.

Graph refinement. If two apparent channels share latent state, naive edge-wise additivity can double-count the same information. The remedy is to refine the time-unrolled DAG so that the shared state appears as an explicit node or edge family, then recompute the min-cut on the refined graph. This usually tightens the certificate more reliably than trying to compensate with ad hoc weight corrections.

Reporting policy. For review and deployment, report a ladder of budgets: worst-case structural, empirical-marginal, and empirical-conditional. The first is the theorem-backed safety baseline. The latter two are optional tightening modes and should be reported only when the underlying measurements are defensible.